

Bidirectional Vietnamese Dialect Transfer

Region-Conditioned Standard→Dialect Generation
with Multi-Task Seq2Seq and Sampling Back-Translation

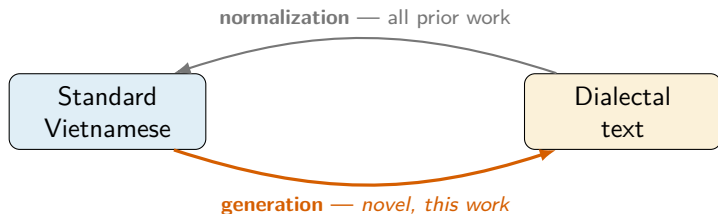
Capstone — IT4772E Natural Language Processing

Hanoi University of Science and Technology

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The problem

- Vietnamese has three dialect regions: **Northern**, **Central**, **Southern**.
- Example (Central): “Anh đi **mô** rồi?” = standard “Anh đi **đâu** rồi?” (“Where did he go?”).



Our goal

One model, both directions *and* three regions — normalization **and** the novel region-conditioned dialect *generation*.

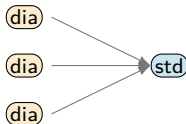
Contributions

1. **Novel task:** region-conditioned standard→dialect generation (Feature C) for all three regions.
2. **Unified multi-task model:** one BART_{pho-syllable} does (A) dialect→std, (B) lexical normalization, (C) std→dialect — selected by a text task prefix.
3. **Evaluation suite for one-to-many:** Dialect Feature Recall, marker-independent Dialectal Edit-Recall, region accuracy, copy-rate (BLEU alone is misleading here).
4. **Data strategy:** oversampling + sampling back-translation for the data-starved regions.

Why is generation harder than normalization?

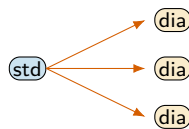
Dialect $\rightarrow$ Standard (A)

- Many-to-one (converging)
- One correct answer
- BLEU is meaningful



Standard $\rightarrow$ Dialect (C)

- **One-to-many** (diverging)
- Many valid realizations
- **Copying scores high BLEU** but does nothing



Consequence

We need metrics that ask *“was the right kind of change made?”*, not *“does it match one reference?”*

Background: conditional sequence-to-sequence

All three tasks are one conditional generation problem. Source x , condition c (the task/region prefix), target y :

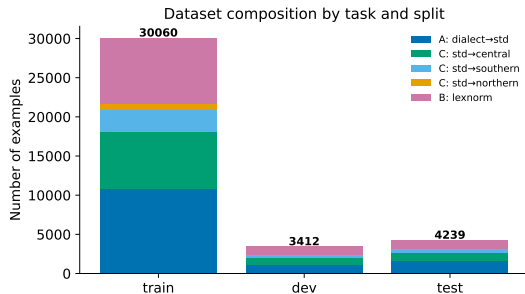
$$P_{\theta}(y \mid x, c) = \prod_{t=1}^n P_{\theta}(y_t \mid y_{<t}, x, c).$$

- **Textual conditioning**: input string is {task}: {source} — no new heads or embeddings.
- **Training**: teacher forcing + token cross-entropy, with **label smoothing** (several outputs are acceptable in one-to-many).
- **Decoding**: *beam search* for the deterministic normalization direction; *sampling* for generation *and* for back-translation (avoids over-regular, mode-collapsed outputs — Edunov et al. 2018).

Data

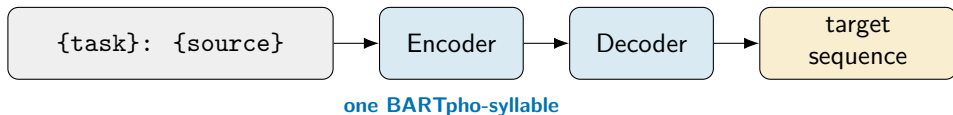
- **ViDia2Std** (Ta et al., AAAI-26): 13,657 dialect/standard pairs, 63 provinces, region-labelled. Used forward (A) *and* reversed (C).
- **ViLexNorm** (EACL 2024): 10,467 teencode→standard pairs (B).
- **Severe imbalance** in ViDia2Std drives the data strategy.

Task	Train	Dev	Test
dialect2std (A)	10,844	1,181	1,597
std2d_central (C)	7,196	767	1,066
std2d_southern (C)	2,893	319	330
std2d_northern (C)	755	95	201
lexnorm (B)	8,372	1,050	1,045
Total	30,060	3,412	4,239



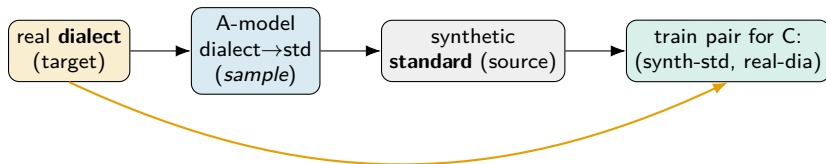
Five task prefixes, one JSON schema:
{task, region, source, target}.

Method (1): multi-task task-prefix model



- One model, **no new heads**; the prefix steers generation, so all tasks share parameters and tokenizer.
- **BARTpho-syllable**: syllable tokenization is robust to dialect/teencode OOV (no fragile word segmenter).
- Prefixes: `dialect2std`, `std2dialect_{northern,central,southern}`, `lexnorm`.
- Expect **positive transfer** among the three noisy-input tasks (both dialect directions + `lexnorm`).

Method (2): reverse corpus + sampling back-translation



- **Reverse corpus:** swap (dialect, standard) $\rightarrow$ (standard, dialect) + region prefix. Target side stays *human-authored*.
- **Sampling back-translation** (Edunov et al. 2018): keep *real dialect* as target, *sample* synthetic standard sources from the strong A-model — sampling, not beam, to avoid mode collapse.
- **Rebalancing:** oversample Northern/Southern; light quality filter (round-trip, length ratio, ≥ 1 marker).

Method (3): transfer-aware model selection

Problem

Pooled validation BLEU rewards a *copying* checkpoint — copying scores high on overlap for std→dialect.

Fix

Select on a **macro per-task BLEU + Dialectal Edit-Recall** composite, so a checkpoint is preferred only if it actually performs the dialectal substitutions.

Evaluation metrics

- **Overlap/semantic:** BLEU, chrF, ROUGE-L, METEOR, BERTScore; **normalization:** WER/CER (A), ERR (B).
- **Dialect Feature Recall (DFR)** — fraction of the reference's regional markers reproduced:

$$\text{DFR} = \frac{|\{m \in R : m \in \hat{y}\}|}{|R|}, \quad R = \{m \in M_r : m \in y\}.$$

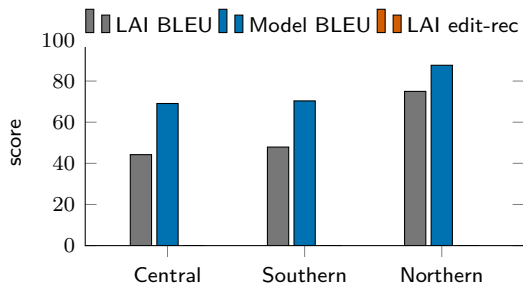
- **Dialectal Edit-Recall** (marker-independent) — did the model make the gold's token changes?

$$\text{Recall} = \frac{|A_{\text{gold}} \cap A_{\text{sys}}|}{|A_{\text{gold}}|}, \quad A_{\bullet} = T(\cdot) \setminus T(x).$$

Copy $\Rightarrow$ Edit-Recall = 0, regardless of BLEU.

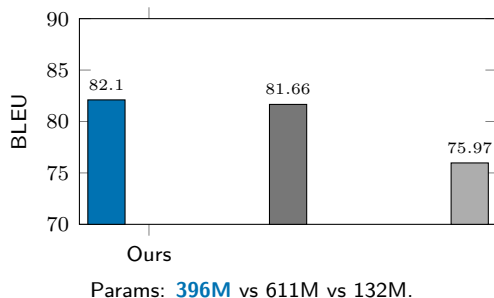
- **Region accuracy + copy-rate:** controllability and under-generation.

Baselines: the copy trap



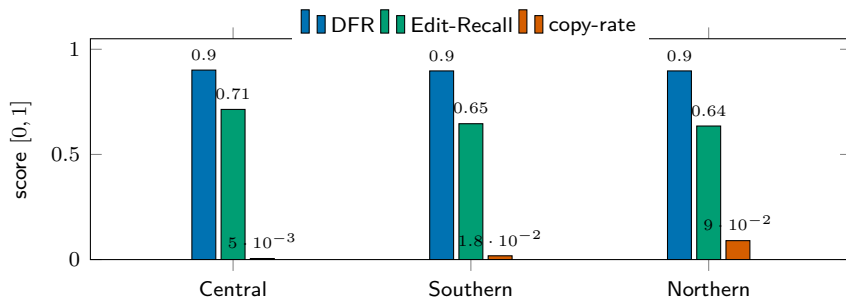
- **Leave-As-Is (copy)** posts high BLEU (74.98 Northern!) but Edit-Recall = 0 — the **copy floor**.
- Dictionary transfers hard (copy-rate 0) but BLEU < 6.
- Rule-based: strongest non-neural on overlap, still recalls only 0.10–0.29 of gold edits.
- Our model **beats every baseline** on Edit-Recall/DFR *and* clears the copy BLEU floor.

Results: normalization (Feature A)



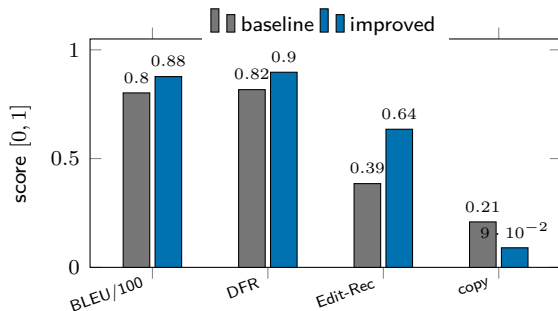
- Our BARTpho-syllable (396M): BLEU **82.10**, WER **0.112**, CER **0.069**.
 - ViDia2Std best (mBART-large-50, 611M): BLEU 81.66, WER 0.123, CER 0.075.
 - ViDia2Std BARTpho-syllable-*base* (132M): BLEU 75.97.
- ⇒ **Matches and edges** the larger SOTA model with $\sim$ a third fewer parameters.

Results: dialect generation (Feature C)



- Overall DFR ≈ 0.90 , Edit-Recall $0.64\text{--}0.71$, copy-rate far below the copy floor of 1.0.

Improvement on data-starved Northern (before → after)



- Northern is the region the data strategy targeted.
- Edit-Recall **0.385→0.635**, DFR **0.817→0.897**, BLEU **80.2→87.7**.
- Copy-rate **falls** 0.209→0.090 — less copying, more transfer.
- Central/Southern maintained; Feature A preserved (BLEU 81.68→82.10).

Analysis: learned lexical substitutions

The model *induced* these from data — not hand-written rules. All are well-attested

regional forms.	Central	Northern	Southern
	này → ni	nhỏ → nhỏ/nhỏ	về → đũa
	vậy → rứa	trời → giờ	không → hông
	đâu/nào → mô	chưa → chữa	thật → thiệt
	gì → chi	giàu → giàu	xem → coi
	sao → răng	mẹ → u / bu	đắt → mắc
	tôi → tui	nhỏ → nhờ	chăn → mền

- Recovers canonical function words, pronouns, and kinship terms per region — evidence of **genuine dialect lexicon**, not surface noise.

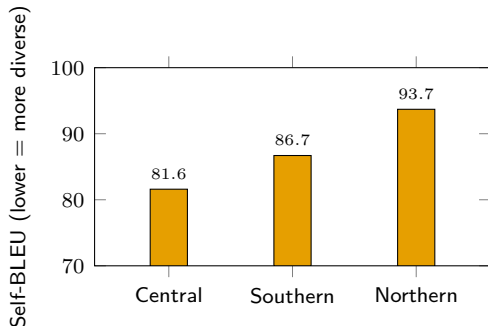
Region controllability: confusion matrix

Rows: *requested* region Columns: region *detected* by the marker classifier.

	North.	Central	South.	none
North.	91	79	16	15
Central	167	850	39	10
South.	98	96	126	10

- Central well controlled (850 on diagonal). **Off-diagonal mass is largely an artefact** of overlapping marker inventories (m , tui span regions) and Northern's orthographic closeness to standard — so we report this matrix + marker-independent Edit-Recall, not a raw error count.

One-to-many diversity under sampling



- Sampling yields **genuine variation**, most for Central, least for Northern (closest to standard, scarcest data).

Standard input

“Tại sao bạn không trả lời tin nhắn của tôi?”

Central samples

“Mắc mớ gì...” · “Răng bạn...” · “Mắc chi...”

Distinct, idiomatic realizations a single-reference BLEU would penalize.

Why syllable tokenization: a probe

Surface form	SentencePiece pieces
“Anh ấy đi đâu rồi?” (std)	_Anh _ấy _đi _đâu _rồi ?
“Ảnh đi mô rồi?” (dialect)	_Ảnh _đi _mô _rồi ?
“Bà ấy nói gì vậy?” (std)	_Bà _ấy _nói _gì _vậy ?
“Bả nói chi rứa?” (dialect)	_B ả _nói _chi _r ử a ?
“Không biết” (std)	_Không _biết
“Hông biết” (dialect)	_H ông _biết

- Standard syllables are **whole tokens**; rarer dialect forms **degrade gracefully** into shorter subwords ($B \rightarrow [B, ả]$, $ra \rightarrow [r, ử, a]$).
- No hard segmentation failure on OOV dialect/teencode — exactly why we pick BARTpho-syllable.

Qualitative: Northern, baseline vs improved

Standard input	Baseline (old)	Improved (new)
Tại sao bạn không trả lời tin nhắn của tôi?	Mắc gì bạn không trả lời...	Tại sao bạn không <i>giả nhời</i> ...
Mẹ tôi nấu ăn rất ngon. Hôm nay trời đẹp quá...	<i>Bu</i> tôi nấu ăn rất ngon. ...(copied)	<i>u</i> tôi nấu ăn rất ngon. ...(still copied)

- Improved model introduces Northern forms (*giả nhời*, *u*) where the baseline copied.
- Honest note: Northern stays hardest — some inputs still copied (the 0.090 residual copy-rate).

Analysis & error taxonomy

- **Under-transfer / copying** (copy-rate, Edit-Recall) — worst on Northern.
- **Wrong-region leakage** (region accuracy / confusion matrix) — markers of another region appear.
- **Over-transfer** (Edit-Precision) — spurious markers.
- **Semantic drift** (BERTScore / human eval).
- One-to-many **diversity**: sampling yields varied valid realizations (Self-BLEU: Central 81.6, Southern 86.7, Northern 93.7; lower = more diverse).

Limitations & future work

- Single dialectal reference per sentence; 63 provinces collapsed to 3 regions.
- DFR depends on a marker list (mitigated by marker-independent Edit-Recall).
- Northern remains smallest; back-translation sources are synthetic.
- Future: human evaluation at scale, ViT5 comparison, iterative back-translation rounds.

Conclusion

- First **region-conditioned standard**→**dialect** generator for Vietnamese.
- One multi-task model: normalization *and* generation *and* lexical normalization.
- Normalization competitive with a much larger SOTA model.
- New transfer-aware metrics expose what BLEU hides; data strategy fixes regional imbalance.

Thank you!

Questions?